INCH-POUND

MIL-S-44090C(GL)
7 November 1990
SUPERSEDING
MIL-S-44090B(GL)
29 March 1988

MILITARY SPECIFICATION

SHIRT, WOMAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a polyester/cotton broadcloth durable press shirt.

1.2 Classification. The shirt shall be of the following sizes as specified (see 6.2).

Schedule of sizes

Short	Regular	Long	X-Long
2	2	2	
4	4	4	
6	6	6	
8	8	8	8
10	10	10	10
12	12	12	12
14	14	14	14
16	16	16	16
18	18	18	
20	20	20	

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8410

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	-	Bag, Plastic, Folded Garment
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester-Covered
V-B-871	-	Button, Sewing Hole, and Button, Staple, (Plastic)
DDD-L-20	-	Label: For Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-F-21840	-	Fastener Tape, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43594	-	Cloth, Interlining, Polyester
MIL-T-43624	-	Thread, Polyester, Spun
MIL-C-43992	-	Cloth, Broadcloth, Polyester/Cotton, Durable Press
MIL-T-44193	-	Thread, Polyester, Textured
MIL-C-44296	-	Cloth, Fusibles
MIL-P-55010	-	Plastic Sheet, Polyethylene Terephthalate

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Technical Manual of the American Association of Textile Chemists and Colorists

Method 88B (Two photographs) - Photographic Comparative Rating of Single and Double Needle Seams

Method 124-82 - Appearance of Durable Press Fabrics after Repeated Home Launderings

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, Research Triangle Park, P.O. Box 12215, Durham, NC 27709-2215.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Type number 304 - Stainless and Heat Resistant Steel
Type number 316 - Stainless and Heat Resistant Steel

(Application for copies of Steel Product Manual should be addressed to the American Iron and Steel Institute, 150 East 42nd St., New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 204 - Methods of Testing Sewing Threads
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standards Shades for Sewing Threads

Department of Defense Standard Shades for Buttons

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the current version before use.

(Application for color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic shirt material shall be polyester/cotton broadcloth, Army Green 415 conforming to MIL-C-43992.

3.3.1.1 Button stay. The button stay material shall be polyester/cotton broadcloth, Army Green 415, conforming to MIL-C-43992 or as an alternate, a commercial button stay material may be used.

3.3.2 Interlinings.

3.3.2.1 Front facings. The interlining for the front facings shall be polyester fiber cloth conforming to MIL-C-43594 except the minimum weight per sq. yd. shall be 2.5 ounces and the minimum yarns per inch shall be 70 warp and 50 filling. The maximum stiffness shall be 0.004 inch pounds in the warp direction and 0.002 inch pounds in the filling direction. The breaking strength requirements shall not apply.

3.3.2.2 Fusible interlining.

3.3.2.2.1 Shoulder loops. The fusible interlining material for the shoulder loops shall conform to type V, style B, class 2 of MIL-C-44296.

3.3.2.2.2 Collar. The fusible interlining for the top collar shall conform to type I, style D, class 3 of MIL-C-44296. As an alternate the fusible interlining as described in 3.3.2.2.1 may be used.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the current version before use.

3.3.2.2.3 Bonding strength. When tested as specified the fusibles in 3.3.2.2.1 and 3.3.2.2.2 shall meet the initial and laundered bonding strength requirements as specified in MIL-C-44296 when fused to the basic material.

3.3.3 Labels. Each shirt shall have a size and a combination identification-instruction label conforming to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

3.3.3.1 Size label. The size label for the shirt shall conform to type IV or VI, class 2.

3.3.3.2 Combination label. The combination identification-instruction label shall conform to type VI, class 15 except that the label cloth shall be dyed Lemon Yellow Cable No. 70205, in conformance with the Standard Color Card of America. The width of the label shall be not more than 1-3/8 inches and the length shall be governed by the contents. The label shall contain the following information:

SHIRT, WOMAN'S, SHORT SLEEVE, POLYESTER/COTTON ARMY GREEN 415, DURABLE PRESS

Washing Instructions

1. Wash with moderately hot water (115°F) using soap or mild detergent.

   a. For machine washing. Use wash and wear cycle.
   b. For hand washing, do not wring or twist.

2. DO NOT STARCH OR BLEACH

3. Dry in tumble dryer at warm setting. Remove immediately at end of drying. Shape and hang.

4. To drip dry, remove from water before last spin cycle. Shape and hang.

5. Touch up is recommended. Iron at a moderate or synthetic material heat setting.

DO NOT REMOVE THIS LABEL

3.3.4 Thread. The thread for seaming, stitching, buttonholes and button sewing shall be cotton- or polyester-covered polyester core, ticket No. 70, 2 or 3 ply and 100, 2 or 3 ply conforming to A-A-50199. The thread shall be stabilized so as not to shrink in boiling water more than 1.5 percent when tested as specified in table IV. Spun polyester thread conforming to class 1 or 2 of MIL-T-43624 may be substituted for the cotton- or polyester-covered polyester core thread as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

Covered polyester	Spun polyester
70, 2 or 3 ply	70, 2 or 3 ply
100, 2 or 3 ply	100, 2 ply

As an option, for all stitching specified in table II, size 100 thread may be used for all operations except for sewing on buttons and making buttonholes. As another option, the looper thread for overedging the front facing fusible interlining to the front edge of the basic material may be polyester textured, type I, class 2, size 18 Tex, conforming to MIL-T-44193.

3.3.4.1 Color and colorfastness. The thread shall be White AH, C.A. 66050. The dyed thread shall show fastness to light, perspiration, and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light, perspiration, and laundering.

3.3.5 Buttons. The buttons shall conform to type II, class K, style 25, 19 line of V-B-871. The color of the buttons shall be shade White BA, C.A. 62031.

3.3.6 Collar stays. The collar stays shall be made from polyethylene terephthalate plastic sheet conforming to MIL-P-55010, except the thickness shall be 0.010 ± 0.001 inch, the color shall be natural (clear or opaque), and the transparency and optical properties shall not apply. Either both ends shall be rounded or pointed with the points rounded, or one end rounded and one end cut at an angle with the point rounded (see 4.4.1.3).

3.3.6.1 Collar stay configuration.

3.3.6.1.1 Unslotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 2-1/4 ± 1/16 inches long.

3.3.6.1.2 Slotted stays. The stays shall measure 3/8 ± 1/32 inch wide and 2-1/4 ± 1/16 inches long. The stay shall have two slots 3/4 ± 1/16 inch long centered along the length of the stay. Each slot shall be 1/32 inch wide and spaced approximately equidistant across the width of the stay.

3.3.7 Fastener tape, nylon. The fastener tape shall be 5/8 inch wide, conforming to type I, class 1 of MIL-F-21840. The color for the neck tab loop fastener tape shall be Gray 3123.

3.4 Design. The shirt is a semi-fitted hip length overblouse style with short sleeves, shoulder loops, convertible collar, six button cut away front, side body, and darts at waist, bust and back neck. The collar and shoulder loops, have a fusible interlining and the collar has stays. Loop fastener tape is attached to front at neck for attachment of a black necktab (see figure 1).

3.5 Figure. Figure 1 is furnished for information purposes only. If there are any inconsistencies between the specification and the figure, the specification shall control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

3.6 Patterns. Standard patterns will be furnished by the Government and, unless otherwise indicated, provide an allowance of 1/4 inch for all seams. In the safety stitch construction the 1/4 inch seam allowance shall be maintained even though the finished seam allowance may measure 3/16 to 7/32 inch due to trimming of the raveled edges of fabric. The standard patterns shall not be altered in any way (except when automated apparel equipment is used for either making or setting shoulder loops see 3.10) and are to be used only as a guide for cutting the contractor's working patterns, which shall be identical to the standard patterns.

3.6.1 List of pattern parts. The components of the shirt shall be cut from material as specified in accordance with the pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature of pattern parts	Cut parts
Basic material	Front	2
	Back	1
	Side body	2
	Sleeve	2
	Shoulder loop	4
	Top collar	1
	Under collar	1
	Stay holder	2
Polyester fiber interlining cloth	Front facing interlining	2
Fusible interlining	Top shoulder loop fusible interlining	2
	Top collar fusible interlining	1
	Marker	
	Collar breakline	

3.7 End item seam appearance, fusible appearance after laundering and adhesion of ultrasonically fused collar stays.

3.7.1 Seam appearance rating. After the finished shirt has been laundered five times (see 4.4.5), the appearance rating of the seams and stitchings shall be as follows:

Part	Minimum average of 3 observations
Fused collar	3.8
Fused shoulder loops	3.8
Sleeve hems	2.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

Armholes	2.8
Bottom hem	2.8
Front-waist darts	2.8
Bust darts	2.8
Back neck darts	2.8
Front side body seam	2.8
Back side body seam	2.8

3.7.2 Appearance of fusibles. After the shirt has been laundered 10 times, it shall show no bubbling or delamination of the fused plies of material when tested as specified in 4.4.5. There shall be no showing of the "strike-through" of the adhesive after fusing.

3.7.3 Ultrasonically fused collar stay resistance to laundering (when applicable). After the shirt has been laundered ten times, the shirt shall show no delamination of the ultrasonically fused collar stays when tested as specified in 4.4.5.

3.8 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.8.1 Stitches, seams, and stitching. Stitch, seam and stitching types specified in table II shall conform to FED-STD-751. Where two or more seam or stitch types are given for the same part of an operation, any one of them may be used. Where stitch typed 401 is used, the looper (under thread) shall be on the inside of the shirt. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained (see 6.5) so that there is no tight or loose stitching. The gage of the safety stitch seams shall be 1/4 inch. The guides and knives on safety stitch machines shall be set to trim only the ravelled ends of the fabric. The edge and top stitching shall be uniformly gaged. Seams required to be worked out, that have a depth between the fold of 1/16 inch or more shall be considered a defect.

3.8.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table II.

3.8.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitchings, shall be back-tacked 1/4 to 1/2 inch. Ends of a continuous line of stitching (except label) shall be overlapped not less than 1/2 inch. The ends of the label stitching shall be overlapped not less than three stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired using 301 stitch type. The series 500 stitch types for sleeve setting seams shall be run off with a 1-1/8 $\pm$ 1/8 inch tail.

3.8.4 Buttonholes. The buttonholes shall be straight-cut, purl or whip type and the finished inside cut length shall be 1/2 to 9/16 inch long. The stitching shall be securely caught in the fabric and the ends shall be securely

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

tacked. When a purl-type buttonhole is used, the purling shall be on the outside of shirt, except for the neck closure buttonhole which shall have the purling on the inside of the shirt. All loose thread ends shall be trimmed or fused to within 1/8 inch of the tack. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

3.8.5 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping or backstitches are used to secure the ends of the stitching.

3.8.6 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

3.8.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the shirts.

3.9 Manufacturing operations requirements. The shirt shall be made in accordance with operation requirements specified in table II. The contractor is not required to follow exact sequence of operations listed unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished shirt.

3.10 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table II, providing that the seam and stitch type are as specified and the finished component conforms to the required size, dimensions, and configurations. When a government furnished shaper pattern is forwarded, caution shall be taken to assure the component conforms to that pattern. Patterns may be altered to accomodate the use of automated apparel equipment as specified in 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut shirts in strict accordance with patterns furnished which show size, directional lines, marks, and notches for proper assembly. Directional lines shall be placed in the warp direction. The use of drill holes are permitted to mark the darts and sewn labels, providing the drill holes are covered by the outer fabrics or label respectively. Otherwise drill holes shall be considered to be holes in the fabric. b. Cut component parts from same piece of the basic material, except the undercollar, and stay holder which may be cut from ends. Parts cut from ends shall approximate the shade of the other parts. c. Cut the fusible and polyester fiber interlinings in accordance with directional lines on patterns. d. Cut button stay from basic material 3/4 ± 1/8 inch diameter or square.						
2.	Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Marking and fusing. a. Mark, ticket or bundle all component parts of the shirt except those parts cut from ends to insure correct size, proper assembly, and uniform shade throughout the shirt. b. Any method of marking may be used except: (1) Metal fastening devices. (2) Sew-on tickets. (3) Adhesive type tickets which leave traces of adhesive on the material after removal of the tickets. c. The use of ink pad numbering machine, rubber stamp or pencil will be acceptable provided the numbers do not show through the outside of the shirt and the numbers are covered by the seam allowance. d. Fusing. For the top collar and top shoulder loops only, position interlining in the center with all edges set back 1/8 inch and fuse.						
4.	Labeling. Center size label (type IV) on topcollar (1/4 inch off center tolerance), so that on the finished shirt, the top of the size label (type IV) will be below the collar breakline stitching (operation 5h.) See operation 17 if type VI label is used.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar.						
	a. Position stay holder to undercollar with side and top edges even.						
	b. Form a tunnel for the collar stays by stitching the stay holder to undercollar with two rows of stitching 7/16 to 1/2 inch apart according to marks on pattern. The collar points shall be aligned with the center of the two rows of stitching and the stitching shall extend to the edges of the stay holder.	301	SSv-2	9-12	70	70	
	c. Insert collar stays into the tunnels with rounded ends abutting the collar and undercollar joining seam and stitch along bottom stay holder from top to bottom edges of collar.	301	SSv-1	9-12	70	70	
	or						
	When using slotted collar stays, position collar stays on undercollar according to marks on pattern so that tapered ends are 3/16 inch back of undercollar point joining seam. Stitch collar stays to undercollar with a 3/4 inch long tack. The stitching shall straddle the center bar of the stay and not penetrate the plastic material.	101		20-24	70		
	or						
	Position collar stays according to marks on pattern so that the tapered ends are 3/16 inch back of undercollar point joining seam. Ultrasonically fuse the collar stay to the undercollar with two rows of fusing points the full length of the stay. Each row shall contain nine to eleven fusing points per inch.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make collar. (cont'd)						
	d. With the collar and undercollar face to face, stitch around side and top edges.	301 or 401	SSe-2(a) SSe-2(a)	9-12 9-12	70 70	70 100	
	e. Trim, turn and press collar points on a collar pointing machine.						
	f. Edge stitch top and sides of collar 1/4 ± 1/16 inch from edge.	301	SSe-2(b)	9-12	70	70	
	g. With collar in flat position, sew a row of stitching forming collar breakline as indicated by the collar breakline stitching marker.	301	SSa-1	9-12	70	70	
6.	Make shoulder loops.						
	Finished appearance. The shoulder loops shall finish smooth and flat and uniform in construction and placement. The buttonholes shall be clean cut and correctly positioned. The shoulder loops shall measure finished 1-5/8 ± 1/8 inches at the narrow end and 1-7/8 ± 1/8 inches at the wide end.						
	a. Position the shoulder loops face to face and stitch around all edges except armhole end.	301 or 401	SSe-2(a) SSe-2(a)	9-12	70 70	70 100	
	b. Turn loop, work out points and edges, and edge stitch along sides and point, 1/4 to 1/16 inch from edge, making sure fused top shoulder loop is on top.	301	SSe-2(b)	9-12	70	70	
	c. Make horizontal buttonhole in the center of each loop 1/8 inch off-center tolerance, with the inside cut end of the buttonhole 3/8 to 7/16 inch from point of loop (see 3.8.4).	301	Whip or purl types	42-52 per buttonhole including tack	70	70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make shoulder loops. (cont'd)						
	d. Press shoulder loops with a hot head pressing machine (see operation 23).						
7.	Attach facing interlining to fronts, and label to facing.						
	a. Position the interlining to the inside of the front facing with edges even.						
	b. Overedge stitch the backside of facing and interlining making certain that the interlining is completely trimmed off during the overedge operation.	502 or 503	SSa-1 SSa-1	6-8 6-8	70 70	70 70	
	c. Position the combination identification and instruction label on the left facing (AS WORN), 3 to 3-1/4 inches from finished bottom and 1 to 1-1/8 inch from finished front, and stitch on all four sides. The stitching shall not be through the printing.	301	LSbj-1	6-8	70	70	
8.	Make sleeves.						
	Join sleeve underseam together with safety stitch.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (overedge)	
9.	Hem sleeve.						
	Turn under bottom edge of sleeve according to notches on pattern with raw edge turned in 1/4 inch and blind-stitch sleeve hem, partially penetrating the sleeve fabric. The finished hem shall be 1-1/4 inches wide.	103 or 106	EFm-1	3-4	70 (Pale Green)		

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Hem sleeve. (cont'd)						
	(NOTE: The hemming of sleeves may be performed before or after joining under arm seams.) When operation 8 is performed after this operation, then the edges of the hem, adjacent to the seam, shall not be staggered. The sleeve seam shall also be bartacked with the seam turned over to one side. The bartack shall be approximately 3/16 inch in length and shall be placed 1/8 inch from bottom of hem across both sides of the seam.	Bartack	---	28 per bartack	70	70	
10.	Make bust, front and waist darts.						
	Finished appearance. The darts shall be properly positioned and finished without pleats, puckers, or gathers.						
	a. Fold and stitch bust darts in fronts tapering to nothing 1/2 to 5/8 inch beyond drill hole. The darts shall be neatly formed and tapered to flat point. The point shall be securely tacked or backstitched.	301	OSf-1	9-12	70	70	
	b. Press bust dart smooth and flat with a heated pressing iron or hot head pressing machine (see operation 23). The edges of the bust dart shall be pressed down.						
	c. Notch corner of front-waist dart and join seams, tapering to nothing 1/2 to 5/8 inch beyond mark for drill hole on pattern.	301	SSa-1	9-12	70	70	
	d. Overedge stitch the front and waist dart seams catching both plies of fabric.	502 or 503	EFd EFd	9-12 9-12	70 70	70 70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make bust, front and waist darts. (cont'd)						
	e. Press front waist darts smooth and flat with a heated pressing iron or hot head pressing machine (see operation 23). The edges of the darts shall be pressed towards the front. Top stitch front waist darts at the curve (where dart goes from vertical towards side seam) 1/16 to 3/32 inch from folded edge. Stitching shall be 1/2 to 5/8 inch in each direction. Back stitching is required.	301	LSbm-3	9-12	70 (Pale Green)	70 Pale Green)	
11.	Make back.						
	a. Fold and stitch back darts tapering to nothing 1/2 to 5/8 inch beyond drill hole. The darts shall be neatly formed and tapered to a flat point. The point shall be securely tacked or backstitched.	301	OSf-1	9-12	70	70	
	b. Press back darts smooth and flat with a heated pressing iron or hot head pressing machine (see operation 23). The folded edges of the darts shall be pressed toward the center back.						
12.	Join front to side body.						
	a. Join side seam of front to front seam of side body with safety stitch, matching notches and easing front to side body between notches in bust area.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (overedge)	
	b. Press seam smooth and flat with a heated pressing iron or hot head pressing machine (see operation 23). The folded edges of the seam shall be pressed toward the front.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Joint shoulder seams and back to side body.						
	a. Join shoulder seams of front and back (top of facing shall not be caught in shoulder seam) with safety stitch extending to overedged stitched edge of facing. Ease back shoulder seam to front shoulder seam between notches and press seam toward back.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (overedge)	
	b. Join back seam of side body to side seam of back with safety stitch matching notches and easing back to side body between notches in shoulder blade area.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 100 (overedge)	
	c. Press seam, smooth and flat with a heated pressing iron or hot head pressing machine (see operation 23). The folded edges of seam shall be pressed toward the back.						
14.	Set shoulder loops.						
	Position shoulder loop so point of loop is 1/4 to 3/8 inch forward of shoulder seam, 3/8 ± 1/16 inch from collar joining finished seam with front edge of loop at armhole opening 1 ± 1/16 inch forward of shoulder seam. Stitch 1/8 to 3/16 inch from raw edge of armhole opening and trim excess even with opening. The excess shoulder loop may be trimmed in operation 15.	301 or 401	SSa-1 SSa-1	9-12 9-12	70 70	70 100	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Set sleeves.						
	Position sleeves in armholes with the front notch matching the front seam of side panel and the back notch matching the back seam of side panel and join with safety stitch catching the raw edges of the shoulder loop in the seaming. The sleevehead fullness shall be distributed evenly between the notches across the top.	515 or 516 or 519	SSa-2	9-12	70 (chain) and 100	100 and 100 (overedge)	
16.	Join cut-away front facing to front together at bottom of overblouse.						
	a. Position the facing and front face to face and stitch angular edges together and continue stitching across width of facing bottom.	301	SSa-1	9-12	70	70	
	b. Turn to finished position and press smooth and flat with seamline to underside.						
17.	Join collar to shirt and attach size label.						
	Finished appearance. The collar shall be carefully joined to the shirt with notches matching without twists, puckers or distorting the collar or shirt. The collar ends shall not be uneven in length by more than 1/8 inch.						
	a. Position size label at center back and stitch 1/8 to 3/16 inch from raw edge of neck area. (Type VI label if applicable).	301	SSa-1	9-12	70	70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Join collar to shirt and attach size label. (cont'd)						
	b. Fold facing back along center front fold line. Insert collar assembly between the front and facing at the collar notch and stitch from folded edge to within 1 ± 1/4 inch from shoulder seam. Notch collar thru front and facing, continue stitching undercollar to neck of shirt.	301	LScf-2(a)	9-12	70	70	
	Turn facings and work out steps. The steps shall finish 7/8 ± 1/16 inch. Turn under unsewn portion of top collar and stitch to neckline 1/32 to 1/16 inch from folded edge, catching the top of the size label in the stitching.	301	LScf-2(b)	9-12	70		
18.	Join shoulder end of facing to shoulder seam.						
	Stitch shoulder of facing to shoulder seam edge, 1/16 to 1/8 inch from edge, for a distance of 3/4 to 1 inch from the outside edge of facing. Facing shall lay smooth and flat without twists or puckers.	301	SSa-1	9-12	70	70	
19.	Hem bottom.						
	a. Turn under bottom raw edge of hem 1/4 inch and stitch 1/16 to 1/8 inch from folded edge.	301	EFa-1	9-12	70	70	
	b. Turn up hem according to notches on pattern and tack back edge of facing to hem not less than 1/2 inch along overedge of facing.	301	SSa-1	9-12	70	70	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Hem bottom. (cont'd)						
	c. Blind stitch hem partially penetrating the body fabric. The stitching shall extend across the front facings, 3/4 to 1 inch from overedge stitching edge. The front side body seams shall turn toward the front and the back side body seams shall turn toward the back. The finished hem shall be 1-1/2 inches wide.	103 or 106	EFm-1	3-4	70 (Pale Green)		
20.	Attach necktab loop fastener tapes.						
	Stitch 5/8 inch square gray loop fastener tape on the right and left fronts with the top edge even with collar joining seam. The front edge of tape shall be positioned 2 ± 1/16 inches from the collar edge of shirt, where collar is joined to neck, for sizes 2 to 12 and 2-3/8 ± 1/16 inches from the collar edge of shirt for sizes 14 to 20.	301	LSbj-1	9-12	70	70	
21.	Make buttonholes.						
	a. Make five vertical 1/2 to 9/16 inch inside cut length buttonholes in the right front of shirt through the facing centered over marks on pattern positioned 3/4 ± 1/16 inch from front edge (see 3.8.4).	301	Whip or purl types	42-52 per buttonhole including tack	70	70	
	b. Make one horizontal 1/2 to 9/16 inch inside cut length buttonhole at top on right front of shirt through the facing with front inner cut edge of buttonhole 3/8 ± 1/16 inch from front edge and 1/4 to 1/16 inch below top of facing (see 3.8.4). The buttonhole shall be parallel to the facing top. Purling on this buttonhole shall be on the inside of the shirt.	304	Whip or purl types	42-52 per buttonhole including tack	70	70	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Clean shirts. a. Before folding and packaging, shirts must have all thread ends trimmed and removed, be free of residual fiber particles, lint and dust, or other foreign substances, and spots and stains removed. b. Remove all shade tickets.						
23.	Pressing. a. As an option to operations 10.b. and e., 12.b. and 13.c., a hot head pressing machine capable of pressing the entire shirt after fabrication may be used (see 6.6). b. Press the bottom of shirt, sleeve hems and under-arm seams. NOTE: The use of steam inflated bag type pressing machine is prohibited. The hot head pressing machine cited on operations 6.d., 10.b. and e., 11.b., 12.b., 13.c., and 25. shall be equipped with controls for accurate temperature (250 to 270°F), time (minimum 10 seconds) and pressure (70 to 80 psi gauge).						
24.	Sew on buttons. Finished appearance. The buttons shall be aligned and securely sewn in the center of the buttonholes ± 1/16 inch on front of shirt and shoulder loops. Collar ends shall be centered equidistant from center of shirt front and shall open not less than 5/8 or						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
24.	Sew on buttons. (cont'd)						
	more than 7/8 inch when collar is buttoned. The lapels shall finish smooth and flat when neck button is unbuttoned and collar is worn open.						
	a. The top button shall be sewn with a back reinforced button stay consisting of 2 plies 3/4 ± 1/8 inch square or diameter of basic material. Position the top button with button stay between left front and facing 1-1/8 ± 1/16 inches from front edge and sew (see 6.8). The stitching shall not penetrate the facing material.	301 or 101		14-16 per button 14-16 per button	70 70	70	
	b. Sew remaining five buttons on the left front 3/4 ± 1/16 inch from the front edge through the front facing.	301 or 101		14-16 per button 14-16 per button	70 70	70	
	c. Sew one button on each shoulder through facing to correspond with the buttonhole in the shoulder loop. The shoulder loop shall lay smooth and flat without puckers or twists when they are buttoned.	301 or 101		14-16 per button 14-16 per button	70 70	70	
25.	Touch-up pressing and closing shirt.						
	a. When required, touch-up fronts, backs, sleeve and bottom hems and collars, before folding for packaging with a heated iron at a permanent press (moderate) setting.						
	b. Button shirt fully.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T23:50Z
Check the source to verify that this is the current version before use.

3.10 Measurements. The finished measurements of the short sleeve shirt shall conform to all the requirements shown in the following table.

TABLE III. Finished measurements (inches)

	2	4	6	8	10	12	14	16	18	20	Tolerance
Collar 1/	12-3/4	13	13-1/4	13-1/2	13-3/4	14	14-1/4	14-1/2	14-3/4	15	± 1/4
1/2 Bust 2/	17	17-1/2	18	18-1/2	19	19-3/4	20-1/2	21-1/4	22-1/4	23-1/4	± 1/2
Cross back 3/	13-3/4	14	14-1/4	14-1/2	14-3/4	15-1/8	15-1/2	15-7/8	16-3/8	16-7/8	± 3/8
1/2 Sweep 4/	17-1/2	18	18-1/2	19	19-1/2	20-1/4	21	21-3/4	22-3/4	23-3/4	± 5/8
Sleeve length 5/:											
Short	8	8-1/8	8-1/4	8-3/8	8-1/2	8-5/8	8-3/4	8-7/8	9	9-1/8	± 1/4
Regular	8-3/8	8-1/2	8-5/8	8-3/4	8-7/8	9	9-1/8	9-1/4	9-3/8	9-1/2	± 1/4
Long	8-7/8	9	9-1/8	9-1/4	9-3/8	9-1/2	9-5/8	9-3/4	9-7/8	10	± 1/4
X-Long	--	--	--	9-3/4	9-7/8	10	10-1/8	10-1/4	--	--	± 1/4
Back length 6/											
Short	19-3/4	20	20-1/4	20-1/2	20-3/4	21	21-1/4	21-1/2	21-3/4	22	± 3/8
Regular	21	21-1/4	21-1/2	21-3/4	22	22-1/4	22-1/2	22-3/4	23	23-1/4	± 3/8
Long	22-1/4	22-1/2	22-3/4	23	23-1/4	23-1/2	23-3/4	24	24-1/4	24-1/2	± 3/8
X-Long	--	--	--	24-1/4	24-1/2	24-3/4	25	25-1/4	--	--	± 3/8

1/ Collar measurement shall be taken along base of collar, from collar edge to collar edge with collar in flat position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

2/ Measure across front at base of armholes with shirt buttoned and laid flat.

3/ At bottom end of darts, measure across back from sleeve setting seam to sleeve setting seam.

4/ Measure along bottom edge from folded edge to folded edge with shirt buttoned.

5/ Sleeve length measurement taken from top of sleeve at shoulder seam along outside edge to bottom of sleeve.

6/ Back length measurements taken along center back from undercollar seam to bottom of shirt.

3.11 Workmanship. The finished shirts shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.3.1 and shall be tested for the characteristics specified in 4.4.3.2.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on the components listed in table IV for the characteristics noted. Unless otherwise indicated, testing shall be conducted in accordance with FED-STD-191. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size and sample size shall be in accordance with the following:

Lot size	Sample size (sample units)
Up to 22,000 (inclusive)	8
22,001 and over	13

The basis for lot size and sample units for testing shall be as follows:

Component	Lot size expressed as	Sample unit for testing
Fusible interlining fused to basic cloth	Yards	1 yard full width
Thread	Cones	1 cone

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method
Fusible interlining fused to basic cloth	Initial adhesion	3.3.2.2.3	4.4.2.1
	Adhesion after 10 launderings	3.3.2.2.3	4.4.2.1 1/
Thread	Shrinkage	3.3.4	ASTM D 204 2/

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

1/ Material shall be laundered as specified in 4.5.1.2 prior to testing for adhesion.

2/ Shrinkage in dry heat shall not apply.

4.4.1.2 Examination of collar stays. Eight collar stays shall be randomly selected from each lot and examined for conformance to configuration and dimensional requirements in 3.3.6. Any nonconforming collar stay found in the sample shall be cause for rejecting the lot represented.

4.4.2 In-process inspection. Visual examination shall be accomplished to ascertain that collar stay tunnels are located and constructed as specified, collar stays are inserted into the collar in the position specified, collar stays are attached and located as specified, and that fusible material is attached to shirt collar and interlining is attached to shirt fronts and shoulder loops as specified. Evidence of nonconformance shall be cause for rejection of all affected items. Final determination of any suspected nonconformance shall be accomplished at the discretion of the Government by partial or complete disassembly of the end items in question.

4.4.2.1 Adhesion testing of fusibles. Prior to fusing each 1,000 parts, a sample of equivalent area to a part shall be fused for visual and adhesion testing. The fusible shall have a 1 inch starter strip or non-fused area at the top portion of the bond strength samples. This can be accomplished either by placing a strip of paper, folding the fusible onto itself or by placing a thin non-adhesive material between the fusible interlining and outershell before fusing. After visual examination for bubbling, delamination, and bleed through, the sample shall be cut to a 1-inch width and peeled in the warp direction using a spring scale or suitable device capable of measuring the peel strength in ounces. Any sample failing the minimum peel strength requirement specified in 3.3.2.2.3 or any bubbling, delamination or bleed-through shall be cause for rejection of the previous 1,000 fused parts. The rejected lot shall not be used in assembly of the shirts. No fusing shall be performed after the rejection of a 1,000 part lot until the machine has been properly adjusted to produce a passing sample. The rejected lot shall then be replaced with a shade matched lot which has been fused correctly. Rejection of the previously fused parts shall not apply to the first sample tested after machine warm-up. A rejected lot may not be used in the assembly of shirts to be presented to the Government. A rejected lot must be disposed of or set aside with clear markings to avoid accidental use. A rejected lot may be replaced with a lot which has been fused correctly and each part must be properly shade matched to the shirt to which it will be joined. A shaded part defect will be scored on all improperly matched shirts.

4.4.3 End item visual examination. The end items shall be examined for defects listed in table V. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects, 6.5 for major and minor A combined defects, and 15 for total (major, minor A, and minor B combined) defects.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

TABLE V. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Pressing	Omitted, poorly pressed, not pressed as specified, or exhibiting a double crease line		201	
	Burned or scorched	101		
	Gloss, wrinkles, or creases			301
	Edge bulky (not pressed solidly)			302
	Sleeve center creased			303
	One or more darts or seams not pressed in the direction specified			304
Material defects and damages	Hole, cut, tear, needle chew, burn, mend, darn, patch, multiple float, coarse or heavy filling, jerked-in-filling, shade bar, or broken or missing yarn:			
	- on inside		202	
	- on outside	102		
	NOTE: The terms outside and inside as used in this document are defined as follows:			
	Outside: Any part of a shirt that is visible when the completely buttoned and closed shirt is examined. This includes either side of the lapel			
	Inside: Any part of shirt that is not visible when the completely buttoned and closed shirt is examined			
	Slub larger than E in all areas, except collar 1/		203	
	Slub larger than C in collar and lapel(s) 1/	103		
	Knot larger than B anywhere on shirt 1/	104		

1/ Knots and slubs. Only knots and slubs on outside that exceed the limits specified in Sears Fabric Defect Scale (see 6.7) shall be scored as defects in evaluating quality of shirts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Shaded parts	Variation in shade within an outside part or between outside parts, except undercollar	105		
	Undercollar badly shaded		204	
Cleanness	Any spot or stain:			
	- on outside	106		
	- on inside		205	
	One or more marking tickets not removed		206	
	Any marking clearly visible on outside	107		
	Thread ends not trimmed to 1/2 inch or less on all seams with the exception of sleeve setting seams, or loose threads not removed		207	
	Visible basting thread not removed		208	
	Series 500 stitch types trimmed to less than 1 inch or more than 1-1/4 inch on sleeve setting operation		209	
Component and assembly	Any component part or required operation omitted or added (unless otherwise classified herein)	108		
	Any operation not performed as specified (unless otherwise classified herein)		210	
Cutting	Any component part not cut in conformance with directional lines indicated on patterns or not cut in accordance with document requirements	109		
Seams and stitching	Accuracy of seaming and stitching:			
	Seam irregular or twisted, puckered 2/, pleated, or wavy	110		
	Part of material caught in unrelated operation or stitching:			
	- caught in up to, and including, 1/4 inch of stitching			305
	- caught in more than 1/4 inch of stitching	111		
	End of stitching when not caught in other seams or stitching backtacked less than 1/4 inch			306

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Stitching overlapping end of thread break less than 1/2 inch			307
	Ends of continuous line of stitching overlapped less than three stitches on labels or less than 1/2 inch on all other continuous stitching			308
	Raw edge more than 1/4 inch long			309
	Gage of stitching and seam allowances:			
	Irregular, i.e., unevenly gaged		211	
	Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			310

2/ Puckering. In the course of the sampling examination, seams suspected of being puckered shall be examined at a distance of 3 feet in comparison with the AATCC photographic comparative rating (see 2.2) for seams. Puckering on a major portion of the suspected seam that equals or is worse than rating 3 for single needle seams shall be scored as a puckered seam.

	Open seam:			
	On all seams except bottom hem and label stitching:			
	Up to and including 1/2 inch			311
	More than 1/2 inch up to and including 3/4 inch		212	
	More than 3/4 inch	112		
	Bottom hem more than 1/2 inch			312
	Label stitching			313
	Runoff:			
	On joining seam, score as open seam			
	Edge or raised stitching (when not caught in an open seam):			
	Up to and including 1/2 inch			314
	More than 1/2 inch		213	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Raw edge:			
	On outside:			
	Up to and including 3/8 inch			315
	More than 3/8 inch		214	
	On inside:			
	More than 2 inches		215	
	1/4 to 2 inches (inclusive)			316
	Seam and stitch type:			
	Not specified seam or stitch type		216	
	Looper thread of 401 stitch type on outside	113		
	Seam and stitch type:			
	Any required line of stitching omitted	114		
	Any line of stitching not beginning or ending where specified (unless otherwise classified herein)		217	
	Any seam not lapped or finished as specified			317
	Broken, missing, or skipped stitches on edge, top, or raised stitching (when seam is seamed, turned, and stitched) including edge, top or raise stitched darts:			
	1/4 to 1/2 inch (inclusive)			318
	More than 1/2 inch		218	
	Stitch tension:			
	Loose tension resulting in a loose seam		219	
	Loose tension on raised or edge stitching resulting in exposed loose thread			319
	Tight tension (stitches break when strain is applied in direction of seam or stitching)	115		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Stitches per inch (to be scored only when the condition exists on more than half the length of seam or stitching):			
	Less than the minimum		220	
	More than the maximum			320
Buttonholes	One or more omitted, added, misplaced, or not finished as specified	116		
	Front end of opening of horizontal top buttonhole not as specified		221	
	Inside cut end of buttonhole less than 3/8 or more than 7/16 inch from point of shoulder loop		222	
	Shoulder loop buttonhole off-center by more than 1/8 inch		223	
	Finished cut length not as specifed		224	
	One or more slanted		225	
	Top bottonhole positioned vertically	117		
	One or more of the lower five button-holes positioned horizontally	118		
	The lower five buttonholes unequally spaced by more than 1/4 inch			321
	The lower five buttonholes less than 11/16 or more than 13/16 inch in from front edge		226	
	Stitches not securely caught in fabric (stitches pull away from fabric)	119		
	One or more broken, or more than two continuous skipped stitches in one or more buttonholes		227	
	Loose stitch tension			322
	Ragged edge in one or more			323
	Not specified type or uncut			324
	Tacking omitted on end		228	
	Purling on top buttonhole (at neck) not on inside (facing side) of shirt	120		
Buttons	One or more buttons missing, broken, or malformed	121		
	Hand sewn button tacked off with less than two stitches			325
	Top button stitching through facing		229	

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Buttons (cont'd)	Buttons not sewn through facing or insecurely sewn		230	
	Stitching not locked at end of cycle (tug at loose end of thread to determine if it will ravel)		231	
	Top shirt button (not collar button) not positioned as specified by pattern marks (See 6.10)	122		
	Not positioned as specified			326
Bartack or tack	Missing	123		
	Insecure, misplaced, not specified size or type, or not serving intended purpose		232	
	Loose stitch tension			327
Labels	Size label:			
	Omitted, incorrect, or illegible	124		
	Off-center by more than 1/4 inch			328
	Top of size label (type IV) on or above collar breakline stitching			329
	Combination identification and instruction label:			
	Omitted, incorrect, or illegible		233	
	Not positioned on lower end of left facing		234	
	Bottom of label positioned less than 3 or more than 3-1/4 inches from finished bottom		235	
	Edge of label less than 1 or more than 1-1/8 inch from finished front			330
Darts	Front or back darts not uniform in length by more than 1/4 inch		236	
	Any dart tapered less than 1/2 or more than 5/8 inch beyond pattern mark			331
	Not properly positioned		237	
	Has pleats, puckers, or gathers		238	
	Point(s) not tacked, backtacked, or backstitched		239	

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Collar	Not uniform in size or shape - any point varying:			
	More than 3/8 inch with corresponding point on matching part	125		
	1/4 to 3/8 (inclusive) inch with corresponding point on matching part		240	
	Collar point uneven in length by:			
	1/8 to 1/4 (inclusive) inch			332
	More than 1/4 and up to 3/8 inch (inclusive) inch		241	
	Over 3/8 inch	126		
	Twisted, not smooth, too full, short, or tight causing collar to turn out	127		
	Collar point not properly forced out	128		
	Collar off-center by 3/:			
	More than 3/16 and up to 1/4 (inclusive) inch			333
	Over 1/4 inch		242	
	Pucker or pleat at neckline		243	
	Edge not worked out (having folds 1/16 inch or more deep)			334
	One or both collar stays missing	129		
	The lapel(s) not finishing smooth and flat when neck button is unbuttoned and collar is worn open		244	
	One or both stays not attached or positioned as specified	130		
	Collar ends open less than 3/8 or more than 5/8 inch when buttoned		245	
	Edge of undercollar exposed beyond edge of top collar:			
	1/16 inch or more for distance of 2 inches or more	131		
	Less than 1/16 inch on major portion of collar		246	

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Collar (cont'd)	Tunnel collar stay stitching missing		247	
	Top and sides of collar not edge stitched as specified		248	
3/ Center of collar is determined by aligning the two shoulder seams at junction of collar.				
Shoulder loop	Not uniform in size or shape - any point varying:			
	More than 3/8 inch with corresponding point on matching part	132		
	3/8 inch or less with corresponding point on matching part		249	
	Not positioned as specified or attached crookedly:			
	Point of loop more than 7/16 inch from collar setting seam		250	
	Front edge of loop at armhole opening less than 15/16 or more than 1-1/16 inches forward of shoulder seam		251	
	Not of the specified dimension	133		
	Loop tight, loose, or badly distorted when buttoned		252	
	Points and edges not worked out. Points not centered as specified			335
	Shoulder area distorted when shoulder loop is buttoned causing fabric pleats or tucks in shirt		253	
Front	Front tight or short, or front facing tight or twisted causing noticeable bulge or twist when shirt is buttoned	134		
	Uneven in length at bottom when buttoned by more than 1/2 inch			336
	Step(s) distorted when neck button (horizontal buttonhole) is unfastened and step is folded back along natural breakline to top and bottom (vertical buttonhole), i.e, when shirt is worn with open neck	135		

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the current version before use.

TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Front facing	Facing not caught as specified			337
	Back side of facing not overedged stitched		254	
	Width of facing not as specified			338
	Shoulder end of facing not attached to shoulder seam edge as specified			339
Armhole and sleeve	Sleeve tight causing pleat or excessive puckers on front, back, or sleeve		255	
	Sleeves reversed		256	
Joining of front to side	Not as specified			340
Joining of back to fronts	Not as specified			341
Bottom hem and sleeve hems	Blindstitching not as specified		257	
	Excessive blindstitch bite			342
	Width of hem not as specified			343
	Hem twisted, pleated, or puckered			344
Neck fastener loop tapes	Less than 9/16 or more than 11/16 inch long			345
	Not specified color			346
	Sizes 2, 4, 6, 8, 10, 12: Positioned less than 2-13/16 or more than 2-15/16 inches from front edge of shirt		258	
	Sizes 14, 16, 18, 20: Positioned less than 3-5/16 or more than 3-7/16 inches from front edge of shirt		259	
Fusible material	Omitted in top collar or shoulder loops	136		
	Delamination:			
	Partial or complete delamination on any fused component	137		
	Bubbling on any fused component	138		
	Any strike through or bleed through	139		
	Any resin transfer to shirt	140		

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension deviating from the specified requirement or sleeves uneven in length by more than 1/2 inch shall be classified as a defect. The lot size shall be expressed in units of shirts. The sample unit shall be one shirt. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested for the characteristics listed in table VI. The lot size shall be expressed in units of shirts. The sample unit shall be two shirts. The sample size shall be as specified in 4.4.1.1. One shirt shall be used to evaluate conformance to the appearance requirements and the other shirt shall be used to evaluate for bubbling and delamination of the collar and delamination of the collar stays. The lot shall be acceptable if one or more sample units fail to meet the requirements for fused collars. Additionally, the lot shall be unacceptable if the lot average (average of all the determinations on the sample units for each characteristic) fails to meet any specified requirement.

TABLE VI. End item tests

Characteristic	Requirement paragraph	Test method
Seam appearance rating	3.7.1	4.5.1.2
Fusibles resistance to laundering	3.7.2	4.5.1.1 1/
Ultrasonically fused collar stay resistance to laundering	3.7.3	4.5.1.1 1/

1/ After laundering and conditioning, the fusible parts shall be visually examined for conformance to 3.7.2 and then, when applicable, the collars shall be opened and the stays shall be examined for conformance to 3.7.3.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the currentversion before use.

Examine	Defect
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of shirts per container is more or less than specified 1/

1/ For this defect, one shipping container in the sample shall be examined

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height, exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Seam appearance rating and fusible resistance to laundering tests.

4.5.1.1 Laundering procedure.

4.5.1.1.1 Apparatus and material used for laundering. The following apparatus and material is required for laundering:

a. Automatic washing machine (see 6.6).
b. Automatic dryer (see 6.6).
c. AATCC Standard Detergent 124 or equivalent.

4.5.1.1.2 Laundering and conditioning. If the shirts or materials selected as test samples weigh less than 4 pounds, ballast to make up a standard 4 pound load shall be added to the test samples. The ballast shall consist of durable press shirts that are approximately the same weight as the test shirts or of pieces of durable press fabric that are similar to the fabric of the test samples. The pieces shall be hemmed to measure 3 feet by 3 feet. The front of

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

all shirts in the 4-pound load shall be unbuttoned. The load shall be placed in the washing machine and 90 grams of detergent shall be added to the washing machine. The washing machine shall be set for 10 minutes on the wash and wear (permanent press) phase, high water level. The washing machine shall be filled with water with a hardness not greater than 50 parts per million (PPM), and the wash water shall be at a temperature of 115° ± 5°F with a cold rinse. Immediately after the washing machine has completed the final spin phase, the entire load shall be removed, untangled, and placed in the dryer. The dryer shall be operated at a temperature of 140° to 160°F until the entire load is dry and then tumbled for an additional 10 minutes with the heat turned off (cool-down phase). The load shall be removed immediately after the machine stops. The above washing and drying procedure constitutes one complete cycle. Test sample material for adhesion testing shall be subjected to ten complete cycles. Test sample shirts for seam appearance rating shall be subjected to five complete cycles. Test sample shirts for collar fusibles and collar stay evaluation shall be subjected to ten complete cycles. After completion of the last cycle and prior to evaluation, the test sample shall be conditioned for 2 hours at 70° ± 10°F and at a relative humidity of 50 to 65 percent. (NOTE: During the interruptions between the cycles and during the conditioning and evaluations, test shirts shall be placed on wooden or plastic hangers and shall be buttoned at the front.)

4.5.1.2 Seam appearance evaluation.

4.5.1.2.1 Apparatus and material used for evaluation. The following apparatus and material is required for evaluation of samples for seam appearance:

a. Overhead lighting procedure apparatus as specified in AATCC Test Method 124-82.
b. AATCC Method 88B photographic standards for evaluating single needle stitching.
c. Evaluation room where the only source of light will be that used for observation. To prevent unwanted reflection of light that will interfere with the observation, the walls of the room may be painted a flat non-reflecting black, or blackout curtains may be hung on each side of the viewing area.

4.5.1.2.2 Evaluation of the appearance of seams. After the test shirts have been laundered and conditioned as specified in 4.5.1.2, the shirts shall be evaluated by three trained observers. The trained observers, independent of the other observers, shall evaluate each of the shirts as follows: The shirt shall be placed on a clothes hanger with the buttons fastened and the hanger suspended on the viewing board in such a manner that the shirt lies flat on the surface of the board. The photographic standards for single needle stitching shall be placed on the viewing board. The observer shall stand directly in front of but 4 feet from the test shirt. The observer shall then compare the test shirt, in each of the areas specified in 3.7, to the photographic standards and shall assign a rating value to each of the specified areas. The rating value assigned to an area by the observer shall be the numerical value of the photographic

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

standard that most nearly matches the area. Seams appearing to fall between the numerical values of the photographic standards may be rated 1.5, 2.5, 3.5, or 4.5. Each rating value assigned by each of the observers shall be recorded. The average rating values assigned by each observer to the same area shall be the rating value for that area. If a rating of the same area made by two observers differs by more than 1.0, the rating of all three observers made for that area shall be discarded and the area shall be re-evaluated by three different trained observers.

4.5.1.2.3 Bubbling and delamination of collars. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2, the collars shall be examined visually for bubbling and delamination.

4.5.1.2.4 Adhesion of ultrasonically fused collar stays. After the test shirts have been laundered and conditioned as specified in 4.5.1.1.2 and collars examined for bubbling and delamination as specified in 4.5.1.2.3. The collar shall be opened and the stays examined visually for delamination and curling.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each fully buttoned shirt shall be neatly folded to measure approximately 13 inches in length by 10 inches in width. The folded shirt shall be secured with stainless steel (AISI type 304 or 316, see 2.2) pins, aluminum clips or plastic fasteners. Each folded shirt shall be unit packed in a close fitting, clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial preservation. The shirts shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Fifty shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and twenty-five in depth. Inside dimensions of each container shall approximate 20-1/2 inches in length, 13-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Fifty shirts of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety CW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each container shall be filled with a box liner conforming to type CF, class domestic, variety DW, grade 200 of PPP-B-636. Level A unit packs shall be packed flat, front up, collars alternated front to rear, two in length, one in width, and twenty-five in depth. Inside dimensions of each container shall approximate 22 inches in length, 14-1/2 inches in width, and 14-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method II as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2) the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Shirts, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), shirts, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 7 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The short sleeve shirts are intended for wear by female military personnel of the Department of the Army.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3 and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.5 Seam appearance. In order to avoid puckering and to produce the flattest seams possible, consideration must be given to the following:

a. The needles used for all seams should be as fine as possible consistent with the size or diameter of the thread used.

b. The throat plate hole should be no larger than necessary to permit free entry of the needle and to prevent flagging.

c. The finest feed (approximately 18 teeth) possible should be used with a pressure foot to exactly match the shape and size of the feed. The face of the pressure foot (underside next to the feed on the machine) should be highly polished to eliminate roughness and reduce drag. Diamond shaped feed teeth have been used quite successfully.

d. The lightest pressure foot pressure should be used to prevent stretching of the top ply of material.

e. In the safety stitch machine (515, 516 and 519), the upper and lower trimming knives should be kept at peak sharpness to prevent "dragging" of the end of the fabric.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

f. In stitch type 301 sewing, the bobbin tension should be adjusted so that when a full bobbin is placed in the bobbin case and threaded through the tension spring, the bobbin and case shall ride slowly down the thread when the free end of the thread is held in the vertical position. Needle tension should be adjusted after the bobbin tension to provide slightly more needle thread than bobbin thread in the stitch.

g. When sewing with stitch type 401 (including the seaming row of the safety stitches), the thread ratio (needle and looper) should be adjusted to produce a stitch with 60 to 70 percent needle thread, and both needle and looper thread tension should be held to the minimum.

h. Thread takeup mechanism should be set on all machines to relieve as much tension as possible when forming the stitch. All thread contacts should be well polished and free of burrs, cuts and rough spots.

6.6 Apparatus. The recommended apparatus for 4.5.1.1 a. and b., and for pressing in operation 23 of table II are:

a. Kenmore Automatic Washing Machine Model 600, or similar washing machine. In case of dispute, the Kenmore Model in use at the procuring agency laboratory.

b. Kenmore Automatic Dryer Model 600, or equivalent.

c. "Formatic" hot head pressing machine or equal. "Formatic" manufactured by the American Laundry Machine Co., Cincinnati, OH.

6.7 Defect scale. Sears Fabric Defect Scales are available from:

"Fabric Defect Replica Kit"
Sears Roebuck and Company
Department 817 (ATTN: BSC 23-29)
Sears Tower
Chicago, IL 60684

6.8 Placement of top shirt button. Distance from center of neck button to center of top shirt button should measure 5-7/8 ± 1/16 inches for size 12 (measurement will change respective to size). Any less results in top shirt button being exposed when worn with the Coat, Classic.

6.9 Subject term (key word) listing.

Clothing, women's
Warm weather

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:50Z
Check the source to verify that this is the current version before use.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8410-A760)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

FIG. 1 - SHIRT, WOMAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

2-1-1531

*U.S. GOVERNMENT PRINTING OFFICE: 1990--504-034/50025

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-44090C(GL)	1990 November 7

3. DOCUMENT TITLE
SHIRT, WOMAN'S, SHORT SLEEVE, POLYESTER/COTTON, ARMY GREEN 415, DURABLE PRESS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 NATICK OP, APR 90 *Previous editions are obsolete.* 198/2

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:50Z
Check the source to verify that this is the currentversion before use.